

Problems on Set

Each problem includes a sample input and expected output for better understanding.

SECTION A - BASIC

Core syntax, built-in operations, and straightforward practice.

1. Write a Python program to create a set and display its elements.

Input	{10, 20, 30, 40}
Expected Output	{10, 20, 30, 40} (order may vary)

2. Write a Python program to add and remove elements from a set.

Input	Set = {10, 20, 30} Add = 40 Remove = 20
Expected Output	{10, 30, 40} (order may vary)

3. Write a Python program to find the union of two sets.

Input	A = {1, 2, 3} B = {3, 4, 5}
Expected Output	{1, 2, 3, 4, 5}

4. Write a Python program to find the intersection of two sets.

Input	A = {1, 2, 3, 4} B = {3, 4, 5, 6}
Expected Output	{3, 4}

5. Write a Python program to find the difference between two sets.

Input	A = {1, 2, 3, 4} B = {3, 4, 5}
Expected Output	A - B = {1, 2}

6. Write a Python program to find the symmetric difference between two sets.

Input	A = {1, 2, 3} B = {3, 4, 5}
Expected Output	{1, 2, 4, 5}

7. Write a Python program to check whether one set is a subset of another set.

Input	A = {1, 2} B = {1, 2, 3, 4}
Expected Output	A is a subset of B = True

Problems on Set

Each problem includes a sample input and expected output for better understanding.

8. Write a Python program to check whether two sets are disjoint.

Input	A = {1, 2, 3} B = {4, 5, 6}
Expected Output	The sets are disjoint = True

9. Write a Python program to remove duplicate elements from a list using a set.

Input	[1, 2, 2, 3, 1, 4]
Expected Output	{1, 2, 3, 4}

10. Write a Python program to find the maximum and minimum values in a set.

Input	{8, 3, 15, 1, 9}
Expected Output	Maximum = 15, Minimum = 1

11. Write a Python program to convert a string into a set of unique characters.

Input	banana
Expected Output	{'b', 'a', 'n'} (order may vary)

12. Write a Python program to count the number of elements in a set.

Input	{10, 20, 30, 40, 50}
Expected Output	Number of elements = 5

13. Write a Python program to clear all elements from a set.

Input	{1, 2, 3, 4}
Expected Output	set()

14. Write a Python program to update a set with elements from another iterable.

Input	Set = {1, 2} Iterable = [3, 4, 5]
Expected Output	{1, 2, 3, 4, 5}

15. Write a Python program to create a frozen set and demonstrate its usage.

Input	[1, 2, 3, 4]
Expected Output	frozenset({1, 2, 3, 4}) Modification is not allowed.

Problems on Set

Each problem includes a sample input and expected output for better understanding.

16. Write a Python program to find common unique words from two sentences using sets.

Input	Sentence 1: python is easy and powerful Sentence 2: python is popular and useful
Expected Output	Common words = {'python', 'is', 'and'} (order may vary)

Problems on Set

Each problem includes a sample input and expected output for better understanding.

SECTION B - INTERMEDIATE

Multi-step logic and practical manipulation of the topic.

1. Write a Python program to find unique vowels present in a sentence using a set.

Input	Education is a powerful tool
Expected Output	Vowels = {'a', 'e', 'i', 'o', 'u'} (order may vary)

2. Write a Python program to find the common elements among three sets.

Input	A = {1, 2, 3, 4} B = {2, 3, 4, 5} C = {3, 4, 6}
Expected Output	Common elements = {3, 4}

3. Write a Python program to find elements that belong to exactly one of two sets.

Input	A = {1, 2, 3} B = {3, 4, 5}
Expected Output	{1, 2, 4, 5}

4. Write a Python program to find missing numbers from 1 to n using set operations.

Input	Set = {1, 2, 4, 5, 7, 10} n = 10
Expected Output	Missing numbers = {3, 6, 8, 9} (order may vary)

5. Write a Python program to calculate the Jaccard similarity of two sets.

Input	A = {1, 2, 3, 4} B = {3, 4, 5, 6}
Expected Output	Jaccard similarity = $2 / 6 = 0.3333$

6. Write a Python program to find duplicate words in a sentence using sets.

Input	python is easy and python is powerful
Expected Output	Duplicate words = {'python', 'is'} (order may vary)

Problems on Set

Each problem includes a sample input and expected output for better understanding.

7. Write a Python program to find all unique non-space characters in a string and display them in sorted order.

Input	data science
Expected Output	Unique characters = ['a', 'c', 'd', 'e', 'i', 'n', 's', 't']

8. Write a Python program to check whether one set is a proper subset of another set.

Input	A = {1, 2} B = {1, 2, 3}
Expected Output	A is a proper subset of B = True

9. Write a Python program to create a set containing squares of all even numbers from 1 to 10.

Input	No user input required.
Expected Output	{4, 16, 36, 64, 100} (order may vary)

10. Write a Python program to find characters that are common to every word in a list.

Input	Words = ['bella', 'label', 'roller']
Expected Output	Common characters = {'e', 'l'} (order may vary)

Problems on Set

Each problem includes a sample input and expected output for better understanding.

SECTION C - ADVANCED

Algorithmic problem solving and deeper application of the topic.

1. Write a Python program to find the longest sequence of consecutive integers using a set for fast lookup.

Input	{100, 4, 200, 1, 3, 2}
Expected Output	Sequence = [1, 2, 3, 4] Length = 4

2. Write a Python program to find all unique pairs whose sum equals a target using a set.

Input	Numbers = [2, 7, 11, 15, 1, 8] Target = 9
Expected Output	Pairs = {(1, 8), (2, 7)} (order may vary)

3. Write a Python program to generate the power set of a set.

Input	Set = {1, 2, 3}
Expected Output	Number of subsets = 8 Subsets include: {}, {1}, {2}, {3}, {1,2}, {1,3}, {2,3}, {1,2,3}

4. Write a Python program to validate that a Sudoku row contains digits 1 to 9 exactly once using a set.

Input	[5, 3, 4, 6, 7, 8, 9, 1, 2]
Expected Output	Valid Sudoku row = True

5. Write a Python program to count distinct elements in every sliding window of size k.

Input	List = [1, 2, 1, 3, 4, 2, 3] k = 4
Expected Output	Distinct counts = [3, 4, 4, 3]

6. Write a Python program to find common neighbors of two nodes in a graph represented with sets.

Input	A: {'B', 'C', 'D'} B: {'A', 'C', 'E'}
Expected Output	Common neighbors of A and B = {'C'}

Problems on Set

Each problem includes a sample input and expected output for better understanding.

7. Write a Python program to recommend friends-of-friends using set operations while excluding existing friends.

Input	A friends = {'B', 'C'} B friends = {'A', 'D'} C friends = {'A', 'D', 'E'}
Expected Output	Recommendations for A = {'D', 'E'} (order may vary)

8. Write a Python program to find the duplicate and missing number from values 1 to n using sets.

Input	List = [1, 2, 2, 4, 5] n = 5
Expected Output	Duplicate = 2 Missing = 3

9. Write a Python program to verify that multiple sets form a disjoint partition of a given universe.

Input	Universe = {1, 2, 3, 4, 5} Groups = [{1, 2}, {3, 4}, {5}]
Expected Output	Valid disjoint partition = True

10. Write a Python program to find the smallest pair of available sets whose union covers the full universe.

Input	Universe = {1, 2, 3, 4, 5} S1 = {1, 2, 3} S2 = {2, 4} S3 = {3, 4, 5}
Expected Output	Smallest covering pair = (S1, S3)